BIOMARKER DISCOVERY & VOICE-BASED CLASSIFICATION OF PARKINSON'S DISEASE: A COMPARATIVE MACHINE LEARNING AND HYPERPARAMETER OPTIMIZATION STUDY

Author: Aaheri Gangopadhyay

Repository: github.com/aaherigangopadhyaystats-code

ABSTRACT

Detecting early signs of Parkinson's disease through non-invasive voice measurements requires more than simple linear screening—it demands capturing subtle, non-linear variations in human speech. In this independent project, I analysed biomedical voice signals from 195 recordings (n = 195) to build a reliable diagnostic model. By testing baseline models like Linear Support Vector Machines (SVM) and Logistic Regression against an optimized Radial Basis Function (RBF) Kernel SVM, I explored how non-linear boundaries handle complex biological feature spaces. While Logistic Regression reached an 82.05% test accuracy and Linear SVM achieved 87.18%, fine-tuning the RBF kernel via 5-fold grid search (C = 100, \gamma = 0.1) bumped performance to 92.31%. Most importantly for medical screening, the final model achieved 100.00% Specificity and 100.00% Precision alongside 90.32% Sensitivity, ensuring healthy individuals were never falsely flagged while maintaining strong detection for true positive cases.

1. PROJECT OVERVIEW & RATIONALE

Voice impairment (dysphonia) is often one of the earliest signs of Parkinson's disease. Small variations in voice pitch, frequency fluctuations (jitter), and volume instability (shimmer) appear long before major motor symptoms show up.

I built this project to answer **two practical questions** using real biomedical voice data:

- Are vocal biomarker patterns straightforward enough to be separated by simple linear boundaries, or do complex non-linear combinations drive a reliable diagnosis?
- How do specific model settings (C and \gamma) impact clinical metrics like false alarm rates versus missed detections?

2. EXPLORING THE DATASET

I worked with the UCI Parkinson's dataset, which includes 195 voice recordings across 22 continuous vocal attributes, plus patient identifiers (name) and diagnosis labels (status: 1 = Parkinson's, 0= Healthy).

Key Vocal Features Analysed:

- Pitch & Frequency: MDVP:Fo(Hz), MDVP:Fhi(Hz), MDVP:Flo(Hz).
- Frequency Fluctuation (Jitter): MDVP:Jitter(\%), MDVP:Jitter(Abs), MDVP:RAP, MDVP:PPQ, Jitter:DDP.
- Amplitude Fluctuation (Shimmer): MDVP:Shimmer, MDVP:Shimmer(dB), Shimmer:APQ3, Shimmer:APQ5, MDVP:APQ, Shimmer:DDA.

- Noise Ratios: NHR (Noise-to-Harmonic Ratio), HNR (Harmonic-to-Noise Ratio).
- Non-Linear Voice Dynamics: RPDE, DFA, spread1, spread2, D2, PPE.

Dataset Breakdown:

- Total Observations: 195 (147 Parkinson's positive [75.38%], 48 Healthy [24.62%])
- Missing Values: None across all 24 columns
- Data Types: 22 continuous numerical features, 1 binary target, 1 string ID

Comparing class averages revealed clear patterns between healthy controls and Parkinson's patients:

- Average Pitch (Fo): Healthy individuals averaged around 181.94 Hz, while Parkinson's patients had a lower average at 145.18Hz.
- Jitter (MDVP:Jitter(\%)): Frequency variation was nearly double in Parkinson's subjects (0.006989) compared to healthy controls (0.003866).
- Harmonic Clarity (HNR): Healthy subjects showed higher vocal clarity (24.63 dB) than Parkinson's subjects (21.82dB).

3. PREPROCESSING PIPELINE

- To keep my evaluation realistic and prevent data leakage:
- Feature Separation: I dropped the string ID column (name) and isolated the 22 continuous voice features from the target (status).
- Train-Test Split: I split the data into an 80\% training set (n = 156) and a 20\% unseen testing set (n = 39) using random_state=2.
- Feature Scaling: Distance-based models like SVMs are sensitive to differences in feature scale (e.g., pitch values in hundreds vs. jitter in decimals). Using StandardScaler, I fitted scaling parameters strictly on the training partition and transformed both sets:

$$X_scaled=(X-\mu_train)/\sigma_train$$

4. MODELING EXPERIMENTS & FINDINGS

I built and compared three distinct classification approaches:

Baseline 1: Logistic Regression

I started with standard Logistic Regression to establish a linear benchmark on scaled features.

- Training Accuracy: 87.18% (136 / 156)
- Test Accuracy: 82.05% (32 / 39)

Baseline 2: Linear Support Vector Machine

Next, I evaluated a Linear SVM to check if a linear decision boundary could better separate the classes.

- Training Accuracy: 88.46% (138 / 156)
- Test Accuracy: 87.18% (34 / 39)

My Observation: While Linear SVM gained 5.13% over Logistic Regression, topping out around 87% suggested that linear models couldn't fully capture the overlapping, non-linear relationships across vocal features.

5. HYPERPARAMETER TUNING (RBF KERNEL SVM)

To handle these complex non-linear patterns, I switched to a Radial Basis Function (RBF) Kernel SVM:

$$K(x, x') = \exp(-\text{gamma} * ||x - x'||^2)$$

I ran a 5-fold cross-validated Grid Search (GridSearchCV) to find the best combination of parameters:

- Regularization (C): [0.1, 1, 10, 100]
- Kernel Scale (gamma): [0.001, 0.01, 0.1, 1,'scale', 'auto']

Grid Search Heatmap Summary (Mean 5-Fold Cross-Validation Accuracy):

C = 0.1 | gamma=0.001: 0.7435 | gamma=0.1: 0.7435 | gamma=1: 0.7435

C = 1.0 | gamma=0.001: 0.7435 | gamma=0.1: 0.8782 | gamma=1: 0.7821

C = 10.0 | gamma=0.001: 0.8464 | gamma=0.1: 0.9292 | gamma=1: 0.8075

C = 100.0 | gamma=0.001: 0.8655 | gamma=0.1: 0.9357 | gamma=1: 0.8075

Best Parameters Selected:

- Optimized Training Accuracy: 100.00% (156 / 156)
- Optimized Test Accuracy:92.31% (36 / 39)

6. DIAGNOSTIC PERFORMANCE EVALUATION

In a clinical screening setting, overall accuracy doesn't tell the full story—minimizing false positives and false negatives is what really matters.

Final Test Set Metrics (n_{test} = 39):

Metric	Score	Practical Significance
Accuracy	92.31%	Overall proportion of correct predictions across test samples.
Sensitivity (Recall)	90.32%	Correctly detected 28 out of 31 actual Parkinson's cases.
Specificity	100.00%	Correctly identified all 8 healthy control subjects with zero false alarms.
Precision (PPV)	100.00%	Every positive flag generated by the model was accurate (28 / 28).

Test Set Confusion Matrix:

	Predicted Healthy (0)	Predicted Parkinson's (1)	
True Healthy	8	0	(Specificity: 100.0%)
True Parkinson's	3	28	(Sensitivity: 90.32%)

Classification Report Summary:

	precision	recall	f1-score	support
Healthy (0)	0.73	1.00	0.84	8
Parkinson's	1.00	0.90	0.95	31
micro avg	0.92	0.92	0.92	39
macro avg	0.86	0.95	0.90	39
weighted avg	0.94	0.92	0.93	39

7. MY TAKEAWAYS & NEXT STEPS

- Non-Linear Dynamics Matter: Moving from a linear model to a tuned RBF kernel boosted test accuracy by +10.26\% over Logistic Regression and +5.13\% over Linear SVM.
- Clinical Reliability: Achieving 100\% Specificity means the system avoids triggering unnecessary anxiety or follow-up testing for healthy individuals, while keeping a strong 90.32\% Sensitivity.
- **Future Improvements:** Moving forward, I want to explore tree-based models like Random Forests and XGBoost alongside SHAP value analysis to break down which specific vocal features (PPE, spread1, or jitter) drive individual diagnostic decisions.